

# On the Safe Road Toward Autonomous Driving

*Phase noise monitoring in radar sensors for functional safety compliance*



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The first approaches to improve vehicle safety were so-called passive safety systems, which did not directly interfere with the driving process but protected the occupants during a crash. In contrast, the first assistance system was the antilock braking system (ABS) successfully introduced in the early 1970s. This active system was developed to avoid an accident by automatically intervening in the braking behavior of the car. At about the same time, the first automotive radar prototype was presented. Since the invention of this very unwieldy radar system, organizations all around the world spent significant efforts in pushing the development of automotive radar systems forward. Today, radar sensors together with ultrasound sensors, lidar, and cameras form the backbone of advanced driver assistant systems (ADASs) as well as autonomous driving (AD), which is in the prototype stage. In particular, because of their robustness against adverse lighting and weather conditions, radar sensors are considered a key technology for modern vehicle safety and comfort systems. Along with the trend toward higher automation, more cars will be equipped with radar sensors in the near future. Because ADASs directly influence the vehicle dynamics, new regulating functional safety (FuSa) requirements, such as the ISO 26262 standard, were introduced. These requirements are mandatory to protect the road users.

Modern automotive radar systems make use of the frequency-modulated continuous wave (FMCW) principle. Despite many advantages to pulse-based radars, one of the most limiting factors of an FMCW radar is the phase noise (PN) contained in the transmit (Tx) signal, which significantly affects the sensitivity and range. To fulfill the ISO 26262 standard, it is thus of high importance to monitor the PN of a radar system throughout its whole lifecycle. In this article, we present the most common PN measurement and estimation techniques for CW signals. Further, we address the problem of estimating the PN of an FMCW signal, which is of particular relevance for automotive FMCW radars and the aforementioned monitoring to fulfill the FuSa requirements. Finally, we present state-of-the-art methods for PN estimation and monitoring in automotive FMCW radar systems.

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## Introduction

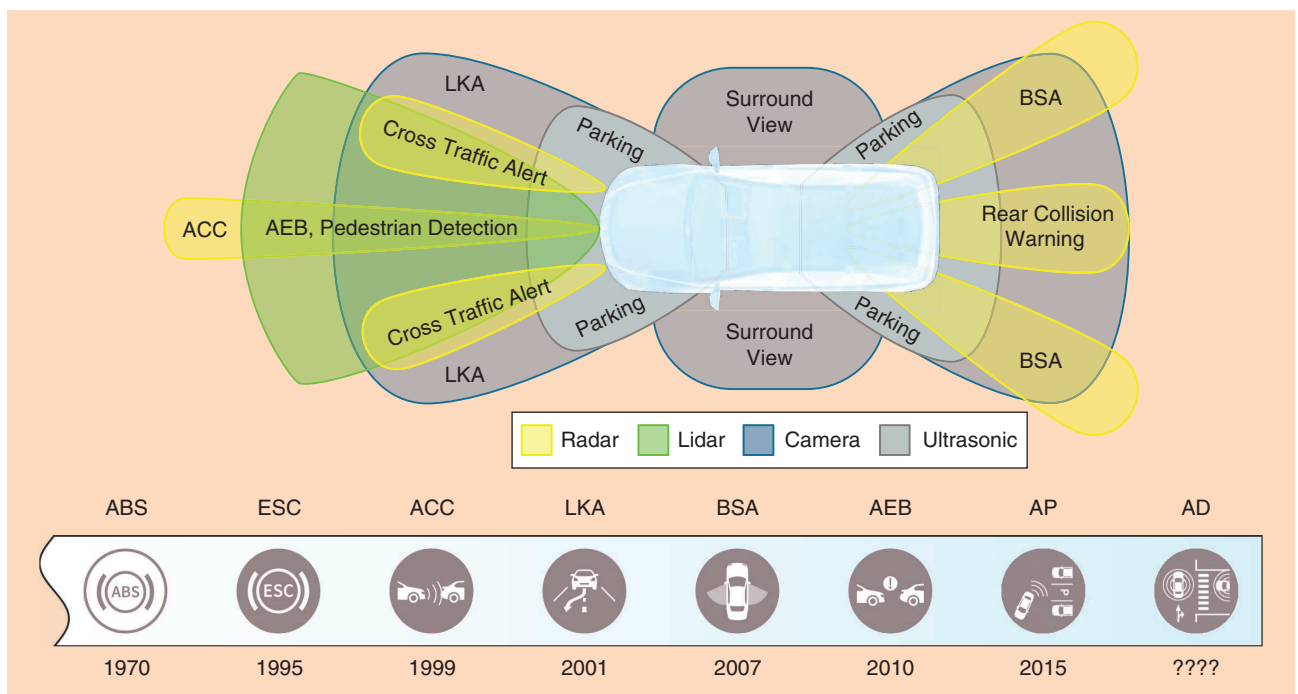
The safety on our roads has been of the utmost importance since the beginning of mass motorization. The first approach to decrease the number of injuries and fatalities on the roads was the development of passive safety features such as seat belts and airbags. The aim of these passive features was to keep the vehicle's occupants protected in case of an accident. Subsequently, active systems for the prevention of accidents were invented. These safety systems proactively intervene into the driving behavior of the vehicle. The forefather of these systems was the ABS, which was followed by the traction control system and electronic stability control (ESC). With a rising degree of integration and a continuous improvement of performance and cost-efficiency, new sensors emerged for the use in ADASs, such as adaptive cruise control (ACC) or autonomous emergency braking (AEB). To fulfill the requirements of these systems, ultrasound sensors, lidars, and cameras as well as automotive radars were introduced [1]. The latter is considered a key technology for the aforementioned applications due to its unique advantage of featuring good performance even in bad weather conditions [2].

The history of ADASs as well as the designated sensor for various driver assistant systems is depicted in Figure 1. With the advent of increasingly complex safety applications, there is a growing demand of so-called imaging radar systems. Particularly for AD, these systems are required as they provide high angular resolution. In general, this is achieved by using a high number of Tx and receive (Rx) channels.

A key challenge for ADASs on the way toward AD is the necessity of safety mechanisms for the detection of systematic and random faults in both hardware and software. That trend is driven

further since the introduction of the ISO 26262 standard in 2011 [3]. Fulfilling this standard requires not only intensive verification and testing of the product functionality but also monitoring of safety-relevant parameters during operation. The latter is typically performed by built-in self-tests (BISTs). An overview of different realizations for possible integrated test solutions is provided in [4], while in [5], a built-in test architecture for radar receiver front ends is presented. These additional building blocks are increasing the complexity of ADASs and AD on the sensor and on the monolithic microwave integrated circuit (MMIC) levels.

Automotive radar applications are typically based on the FMCW principle. These radar systems are capable of simultaneously measuring the distance, velocity, and angle in elevation and azimuth of surrounding objects. One of the most limiting factors of FMCW radars in terms of range precision and detection sensitivity is signal distortions due to the PN contained in the Tx signal [6]. This signal is typically generated by a phase-locked loop (PLL), which inherently produces random fluctuations in the phase of the signal (the PN). Traditionally, the PN of a frequency-generating circuit is described by its power spectral density (PSD) measured after production under the use of expensive measurement equipment. However, the PLL may change its PN behavior with time, temperature, or supply voltage. Therefore, the estimation of the PN PSD is an important monitoring function, which requires on-chip and in-operation estimation techniques. To achieve the FuSa goals for ADASs and AD, it has to be ensured that the radar MMICs do not generate PN out of specification. This can be accomplished with proper monitoring features that are an integral part of this article. Prior to investigating this, the concept of FMCW radars is explained.



**FIGURE 1.** The history of active driver assistance systems, from the introduction of ABS in the early 1970s until future AD, and spatial coverage of ADAS sensors in a modern car. AP: autonomous parking; BSA: blind spot assist; LKA: lane keep assist.

## The FMCW radar principle

Modern automotive radar sensors typically use an FMCW modulation scheme working in frequency ranges from 76 to 81 GHz. Therefore, a frequency-generating circuit, e.g., a PLL, produces a linear frequency ramp (chirp), which is used as a Tx signal. When initially neglecting the effects of the PN and other noise sources, an ideal version of the Tx signal can be written as

$$s_{Tx}(t) = A_{Tx} \cdot \cos(2\pi f_0 t + \pi k t^2), \quad (1)$$

for  $t \in [0, T_{CH}]$ , where  $T_{CH}$  represents the time duration of one chirp-period. In (1),  $A_{Tx}$  and  $f_0$  are the amplitude and the starting frequency of the Tx signal, respectively. Further,  $k = B/T_{CH}$  is the frequency slope of the chirp, whose frequency sweep (often referred to as *sweep* or *chirp bandwidth* in radar literature) is denoted as  $B$ .

The Tx signal  $s_{Tx}(t)$  is emitted by a Tx antenna and reflected by the surrounding objects. For the sake of simplicity, the objects will be represented by simple point targets, each producing a single reflected electromagnetic wave. As illustrated in Figure 2(b), every object reflection can therefore be modeled as a damped and time-delayed version of the Tx signal. The sensed Rx signal can be calculated as the superposition of all object reflections and is thus given by

$$s_{Rx}(t) = \sum_{i=1}^N A_{Tx} \cdot \alpha_i \cdot \cos(2\pi f_0(t - \tau_i) + \pi k(t - \tau_i)^2), \quad (2)$$

where  $\alpha_i$  is a damping factor due to path and reflection losses corresponding to the signal contribution of the  $i$ th object reflection. The time delay  $\tau_i$  is typically referred to as round-trip delay time (RTDT) and represents the propagation time of the electromagnetic wave traveling from the Tx antenna to the object and back to the Rx antenna. Thus, the RTDTs  $\tau_i$  are proportional to the distances to the respective objects  $i = 1, \dots, N$  in the radar channel.

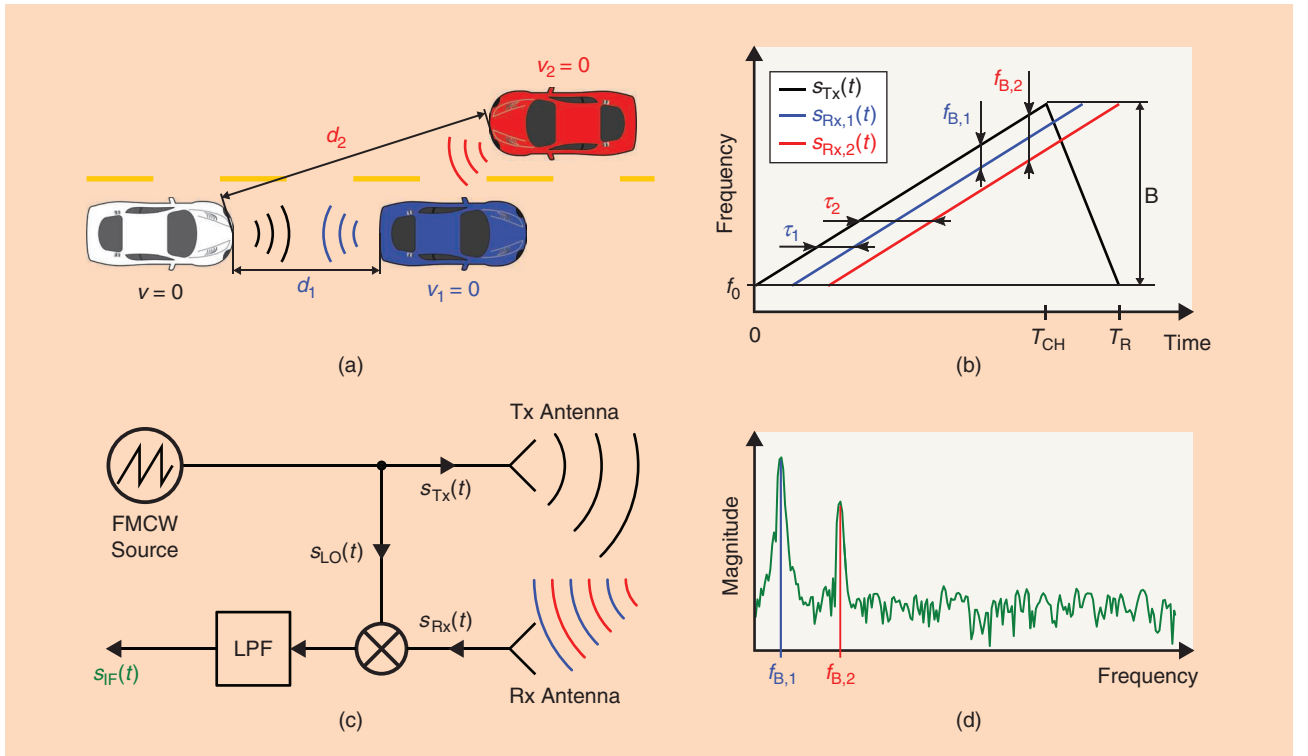
In the receiver, the Rx signal is multiplied with the instantaneous Tx [local oscillator (LO)] signal and subsequently filtered by an analog low-pass filter (LPF). The resulting signal is termed the *intermediate frequency (IF)* signal in the radar literature. A good overview of potential algorithms to process this IF signal to obtain range, relative velocity, and angle information of the surrounding objects is provided in [7]. In the sequel, we will present one possible method using the conventional 3D fast Fourier transform (FFT) approach in tutorial style.

## Range estimation

Assuming the static scenario (i.e., nonmoving objects) as depicted in Figure 2(a), the IF signal is given with

$$s_{IF}(t) = [s_{Tx}(t) \cdot s_{Rx}(t)] * h_L(t) = \sum_{i=1}^N A_{IF,i} \cdot \cos(2\pi f_{B,i} t + \Phi_i), \quad (3)$$

where  $*$  denotes the convolution operator and  $h_L(t)$  is the impulse response of the ideally assumed LPF. Further,  $A_{IF,i} = (A_{Tx}^2 \cdot \alpha_i)/2$  and  $\Phi_i = 2\pi f_0 \tau_i - \pi k \tau_i^2$  are the amplitude and a constant phase term corresponding to the signal



**FIGURE 2.** The FMCW radar system concept and range estimation method using an FFT approach in a static scenario (i.e., nonmoving objects). The (a) scenario, (b) time–frequency domain representation, (c) basic system model, and (d) range spectrum.

contribution of the  $i$ th object reflection, respectively. The term  $f_{B,i}$  is referred to as *beat frequency* and represents the instantaneous frequency difference between the LO and the Rx signal of each point target. As illustrated in Figure 2,  $f_{B,i}$  is proportional to the (one-way) distance  $d_i$  between the radar sensor and the  $i$ th object with

$$f_{B,i} = k \cdot \tau_i = \frac{B}{T_{CH}} \cdot \frac{2d_i}{c_0}. \quad (4)$$

According to (4) the estimation of  $f_{B,i}$  enables the range measurement, which is typically done using an FFT. An exemplary magnitude spectrum of the IF signal is depicted in Figure 2(d) showing two object reflections with the beat frequencies  $f_{B,1}$  and  $f_{B,2}$ , respectively.

### Velocity estimation

We now consider a dynamic scenario with two objects at the same distance but moving in opposite directions with relative velocities  $v_1$  and  $v_2$ , respectively. The scenario is depicted in Figure 3(a). Because the objects are located at the same distance, they share the same range bin in the magnitude spec-

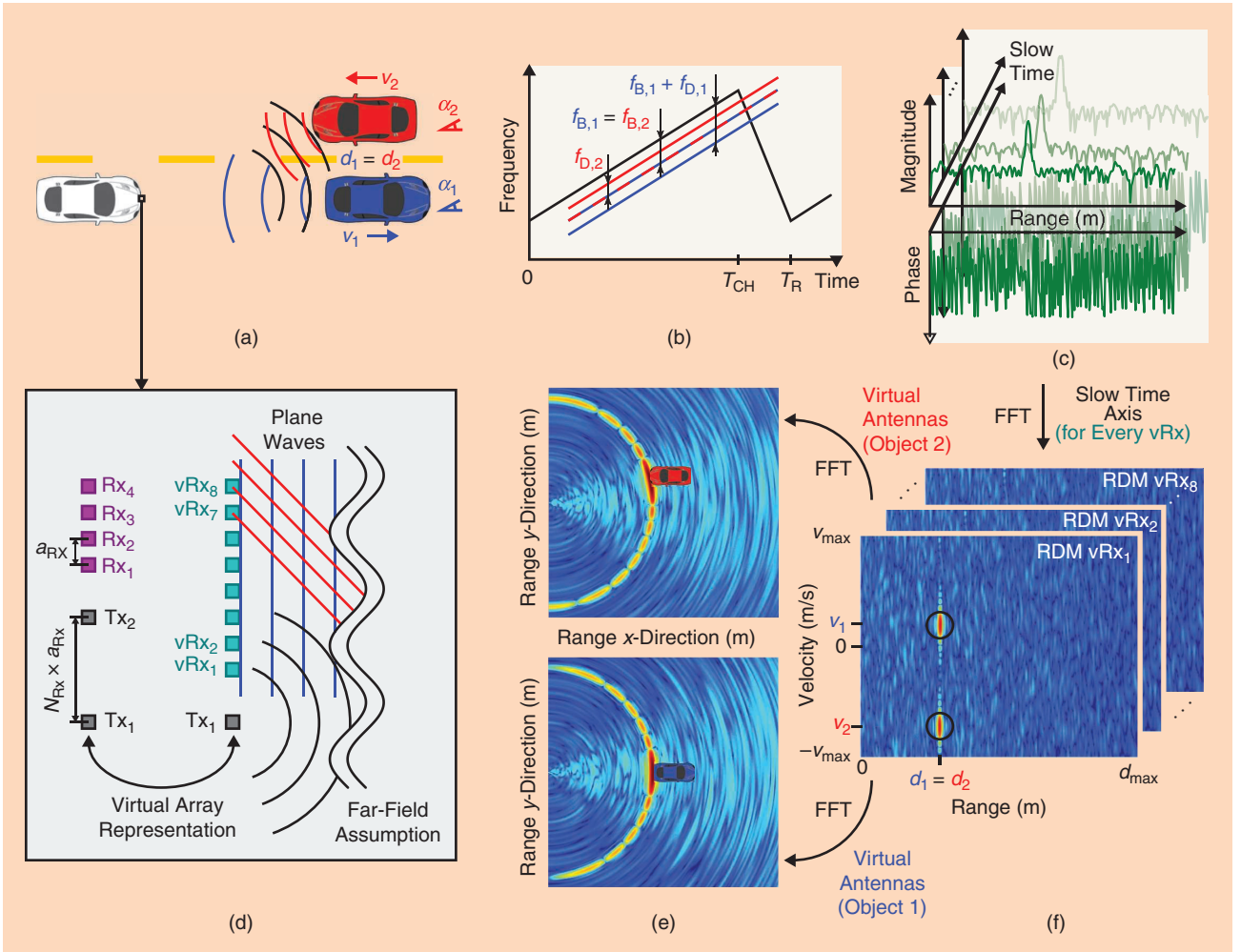
trum and thus cannot be separated in the range domain. To estimate the velocity of the objects, one can take advantage of the Doppler effect. That is, the frequency of the received wave and, therefore,  $\tau_i$  slightly change over time according to the velocity of the object. To be able to measure the Doppler effect, several consecutive ramps are emitted as illustrated in the time–frequency domain representation in Figure 3(b). In a scenario with moving objects, the RTDT can be defined by

$$\tau_i(t) = \tau_{0,i} + \frac{2v_i}{c_0} t \quad (5)$$

with  $\tau_{0,i}$  representing the initial RTDT of the  $i$ th object in the radar channel. Using the time-varying RTDT (5), based on (3), and neglecting the terms providing insignificant impact on the radar measurement, the IF signal is given with

$$s_{IF}(t) = \sum_{i=1}^N A_{IF,i} \cdot \cos(2\pi(f_{B,i} + f_{D,i})t + \Phi_i), \quad (6)$$

containing the Doppler frequency  $f_{D,i} = f_0(2v_i/c_0)$ . To obtain the relative velocity, the Doppler frequencies have to be determined by estimating the phase difference between several



**FIGURE 3.** A schematic representation of how to build up a radar cube consisting of several RDMs using the concept of a virtual antenna array to estimate range, relative velocity, and angle of objects in the surrounding of the radar sensor. The (a) scenario, (b) time–frequency domain representation, (c) range spectra, (d) antenna setting, (e) angle estimation, and (f) radar cube. vRx: virtual Rx antenna.



consecutive ramps. One very common approach for velocity estimation in FMCW radar applications is to calculate the so-called range Doppler map (RDM) [8]. This 2D data set contains the range as well as the velocity information of all objects in the radar sensor's field of view. The RDM is obtained by stacking the complex-valued spectra of the IF signals obtained for the individual ramps into a matrix, and performing an FFT along the professed slow-time axis. The procedure as well as the resulting RDM is depicted in Figure 3(c) and (f). It can be seen that in the range spectrum, the object bins overlap, while in the RDM, they are separable in velocity.

### Angle estimation

To determine the angle of the surrounding objects, it is possible to use superresolution algorithms such as multiple signal classification (MUSIC) [9] or estimation of signal parameters via rotational invariances technique (ESPRIT) [10]. However, because of their high computational complexity, these algorithms are not widely used in automotive applications [7]. Therefore, we present an FFT-based algorithm to estimate the angle of arrival, which is commonly utilized in practice. For this purpose, multiple-input, multiple-output (MIMO) radar systems in combination with time-division multiplexing are used. These systems are composed of several Tx and Rx antennas to estimate the direction of arrival for every detected object. If the spacing between the antennas is chosen as shown in Figure 3(d) (where  $a_{Rx}$  is typically chosen as half the wavelength) with  $N_{Tx} = 2$  Tx antennas and  $N_{Rx} = 4$  Rx antennas, the MIMO radar system imitates an antenna array with one Tx antenna and  $N_{Tx} \times N_{Rx}$  equally spaced Rx antennas. This enlargement of the antenna array is referred to as *virtual array representation*, which increases the spatial resolution of the FFT-based estimation algorithms [7]. Note that this representation of the antenna array is suitable only if the far-field assumption is fulfilled.

To estimate the direction of arrival, the individual complex-valued RDMs of all of the virtual Rx (vRx) antennas are arranged in a 3D matrix, which forms the supposed radar cube. Depending on the angle of incidence, there is a certain phase difference between the sensed signals of the vRx antennas as can be seen in Figure 3(d). This phase difference may be estimated by performing an FFT across the third dimension of the radar cube for every detected object [7]. The results of these final FFTs reveal the angle of the objects in the radar channel and are often depicted as illustrated in Figure 3(e).

### Impact of PN in automotive radar systems

One of the most dominant signal distortions in an FMCW radar sensor is random fluctuations in the phase of the Tx signal (PN). It has a significant impact on the quality of the IF signal and therefore limits the performance of the radar system. The impact of PN on the spatial resolution due to a degrading signal-to-noise ratio as well as a limitation of the maximum range of the radar system is shown in [11]. In addition, according to the simulation results in [12], a deterioration of the PN characteristic may lead to a broadened object peak, which directly affects the range resolution.

To further analyze the impact of PN on the radar measurement, (1) is modified to additionally include the PN according to

$$s_{Tx}(t) = A_{Tx} \cos(2\pi f_0 t + \pi k t^2 + \varphi(t)). \quad (7)$$

On the basis of (7), the distorted IF signal can then be written as

$$s_{IF}(t) = \sum_{i=1}^N A_{IF,i} \cdot \cos(2\pi f_{B,i} t + \Phi_i + \Delta\varphi_i(t)), \quad (8)$$

where  $\Delta\varphi_i(t)$  is the alleged decorrelated PN (DPN) corresponding to the  $i$ th object reflection. It is defined as the difference between the PN contained in the instantaneous Tx signal and the PN delayed by  $\tau_i$ , and thus, it can be written as  $\Delta\varphi_i(t) = \varphi(t) - \varphi(t - \tau_i)$ . Assuming that the PN and the DPN are zero-mean wide-sense stationary ergodic processes, they can be represented by their PSDs [13]. Further applying the concept of the average PSD, it can be shown that the DPN, generated by objects close to the radar system, is raising the overall noise floor [14]. This becomes a problem especially if the radar sensor is mounted behind the vehicle's bumper. Besides the power loss, the bumper also acts as a close object reflecting a high amount of signal power back to the sensor [15]. This unwanted signal reflection in combination with the PN further limits the achievable sensitivity and accuracy of the radar system. One method to mitigate this apparent short-range (SR) leakage caused by the bumper is the integration of an artificial on-chip target (OCT), which essentially consists of a short delay line. The OCT is used to estimate the DPN of the bumper reflection. On the basis of cross correlation properties of the DPN of the OCT and the bumper reflection, almost perfect leakage cancellation is achieved [16]. This method increases the probability to detect objects with a very low radar cross section (RCS) [17]. Particularly, reflections from pedestrians, which would otherwise be covered by the noise floor, may become detectable.

The method proposed in [16] is suitable for objects at an approximately constant and close distance to the sensor, e.g., the bumper. The problem becomes somewhat different if we consider the scenario depicted in Figure 4, where the car is equipped with a radar sensor. The truck in front of the car is located at a distance of 1.5 m and has an RCS that is about a thousand times higher than the RCS of the pedestrian at a distance of 10 m [18], [19]. As shown in the simulated range spectra, the radar system with a PN PSD meeting the requirements is able to detect the hiker next to the road. If the noise figure is increased by 6 dB, the pedestrian is masked by the noise floor and is no longer visible to the radar.

In [20], the authors showed that there may be an additional distortion, affecting the ramp linearity, caused by switching in digital circuits or the transient response of the PLL. These deviations result in sinusoidal nonlinearities in the chirp signal and are referred to as *spurious oscillations* (spurs). In Figure 5, an exemplary PN PSD containing a sinusoidal nonlinearity at  $f_c = 2$  MHz is shown. This spur generates conjectural ghost targets at the frequencies  $f_{GT,1,2} = f_c \pm f_B$ , which will be interpreted as potential objects by the radar sensor.

In this section, we have shown that PN may directly cause personal injuries due to a deterioration of accuracy and detection sensitivity. Because ADASs as well as AD aim to make it less dangerous for humans to participate in traffic, it is of high importance to verify that all elements involved in the radar measurement work properly. A guideline on how the development of such functional safe products can be achieved is provided by the ISO 26262 standard, which will be the topic of the next section.

## FuSa according to ISO 26262

Statistically, an assisting system or an automated driver has to operate a vehicle in a much safer way than a human driver. Consequently, these systems need to meet very high integrity standards and demands toward FuSa. For the automotive industry, ISO 26262 enables standardized development of

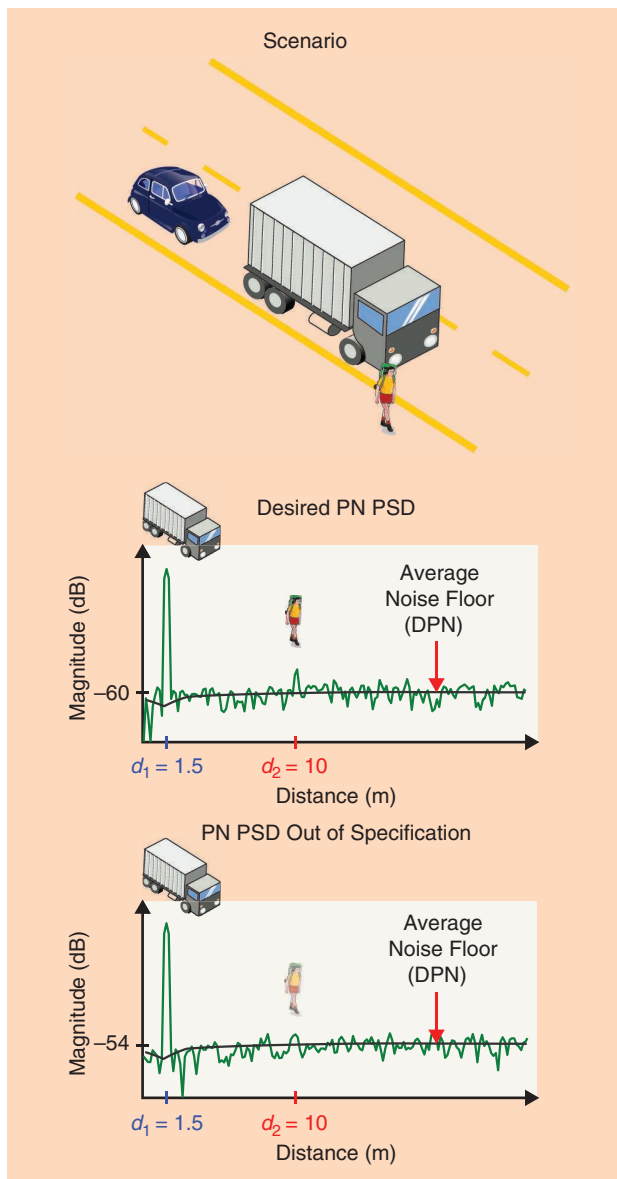
functional safe products [3]. In this article, we glance at some of the technical implications of the ISO 26262 standard for the development of a radar sensor only, leaving aside other activities demanded by the standard, such as planning, change management, competence management, tool qualification, audits and assessments, and also safety analyses.

To avoid unreasonable risks, ISO 26262 guides the development in an attempt to

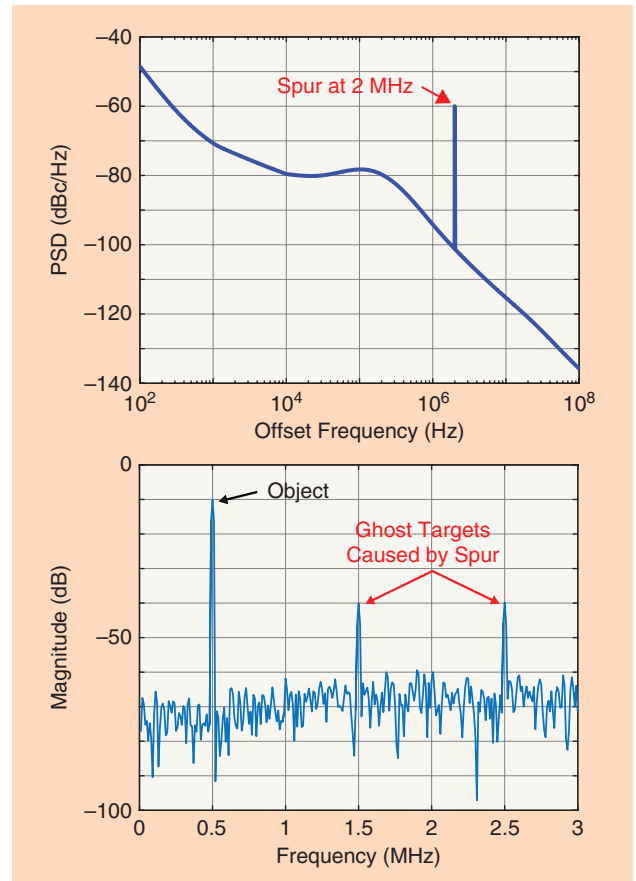
- 1) remove hazards from random hardware faults, which are able to impair the nominal function
- 2) prevent systematic faults in hardware and software that are introduced during development
- 3) evade faulty specification of product properties.

The very high complexity of automotive systems is acknowledged by defining extensions on top of system development according to the V model and requirements engineering. Looking at the left leg of the V, a derivation and refinement process is sketched to break down the high-level goals into smaller technically feasible problems in as many steps as necessary. This could be exemplified by the chosen levels depicted in Figure 6.

The sensor function is described by requirements on the product level (radar sensor). In the elaboration of the product architecture, subparts (sensor elements) as well as the requirements to these subparts are defined. Further down the left leg of



**FIGURE 4.** A scenario where the overall noise floor of a radar measurement is increased by an object close to the radar sensor, and the corresponding simulated range spectra with good and poor PN characteristic.



**FIGURE 5.** An exemplary PN PSD containing a sinusoidal nonlinearity at 2 MHz as well as the range spectrum with an object at  $f_B = 0.5$  MHz and two ghost targets at  $f = 2 \text{ MHz} \pm f_B$ .

the V, this derivation and refinement process can be continued to sub-subparts (hardware/software) and further, if necessary.

In requirements engineering, ISO 26262 becomes directly apparent because it adds an additional safety layer, the safety requirements, alongside the usual technical requirements. The relevance of the safety requirements is affirmed by the assignment of an automotive safety integrity level (ASIL). The integrity level can be interpreted as the degree of certainty needed that the unit fulfills this requirement. With respect to requirements without an ASIL, i.e., all other regular requirements, ISO 26262 is very conservative by regarding them as systematically violated, which implies the assumption that a fulfillment cannot be relied on at all. From this, it follows directly that if a fault of those technical requirements in subparts or sub-subparts can lead to a violation of a safety requirement on the product level, their fulfillment needs to be checked and monitored by a safety mechanism. This needs to be done continuously and, especially after all final production tests, during regular operation in the field.

On the one hand, broad ASIL allocation can enhance product safety. On the other hand, an economic product development aims at having as few ASIL assignments as safely possible. This is because design and verification efforts of parts and requirements with ASIL are drastically increased to meet the higher integrity needs. Furthermore, the standard defines the inheritance of the ASIL by subparts of the product. Hence, it is reasonable to ensure the safety on higher levels to be able to develop the subparts without an ASIL.

Let us consider an emergency braking system to demonstrate the implications of ISO 26262. For an emergency braking system, it could be required with ASIL to take action if a pedestrian is directly in front of the car. Another safety requirement could be that false emergency brakings must not occur. Both of these safety requirements on the system level potentially lead to safety requirements at the product level of the radar sensor regarding a certain resolution or PN level.

As a consequence, the fulfillment of these requirements could be ensured by ASIL assignments and/or safety mechanisms for all contributing subparts and sub-subparts. However, this comes at an enormous development effort. A much more economic and elegant way is to monitor the PN PSD on the product level using

additional BIST architectures in combination with digital signal processing like the methods presented in the “PN Estimation of FMCW Signals” and “PN Estimation With FMCW Signals in Cascaded System Configuration” sections.

## PN measurement of CW signals

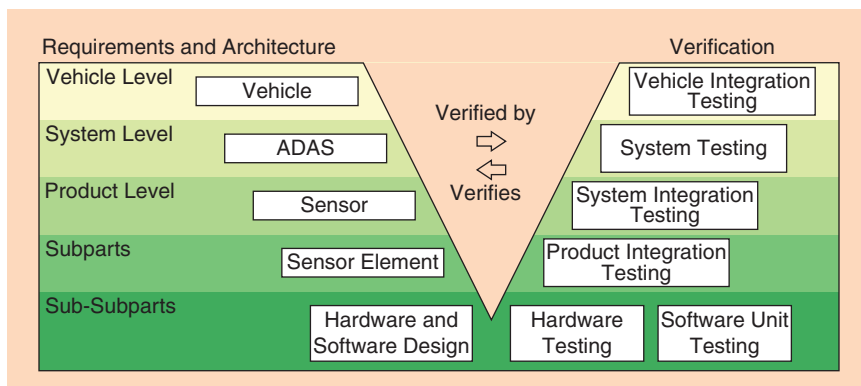
The characterization of a radar MMIC is usually done after production to ensure the circuit operates within its specified performance parameters. As part of this characterization, the PN PSD is measured with a spectrum analyzer in the lab. This assessment is crucial because PN is one of the most meaningful figures of merit of a PLL.

There are several methods to measure or estimate the PN PSD, each satisfying various requirements. Typically, the signal, distorted by PN, is converted to baseband to further extract either the phase or frequency information of the carrier. Thereafter, a frequency selective measuring device is used to measure the noise as a function of the offset frequency.

The common way to verify the specifications of the PLL of FMCW radar MMICs is to use the direct spectrum technique. This simple yet effective method directly measures the PN PSD of the PLL, utilizing the LO of the spectrum analyzer for down-conversion. In general, however, a direct-spectrum reading has a disadvantage [21]. A spectrum analyzer is generally not able to differentiate between amplitude noise and PN, but because the effect of amplitude noise is negligible compared to that of PN, it does not pose a problem for our application at hand [13].

As previously mentioned, the PN behavior may change due to voltage and temperature variations as well as aging. Therefore, it is necessary to estimate the PN PSD with an on-chip circuitry. This enables monitoring of the PN PSD throughout the whole lifecycle of the radar MMIC. Most state-of-the-art on-chip PN measurement techniques for CW signals use the delay line discriminator (DLD) approach [21]. In contrast to the direct spectrum technique, this method does not require a reference clock to convert the signal of interest into baseband. In Figure 7(a), the block diagram of a DLD-based measurement system is shown. To perform the PN measurement, the signal of interest is split into two paths. One signal is directly passed to the mixer, which acts as a phase detector, while the other one is sent through a

delay line first. To remove the resulting sum frequency, the output of the phase detector is put through the LPF. Because of the fact that the involved signals have identical frequencies, the phase detector output is essentially a fluctuating voltage related to phase fluctuations of the PLL. Please note that the signals need to be at a 90° phase shift during the whole measurement to achieve an accurate PN value. Therefore, the authors of [22] presented an on-chip PN measurement circuit including a feedback loop to calibrate the delay line, which ensures perfect quadrature conditions and improves



**FIGURE 6.** A system development process according to the V model.

the PN sensitivity. A different concept to enhance the PN sensitivity is to use an injection-locked-oscillator phase discriminator as proposed in [23].

Unfortunately, the techniques described previously constrain the input signal to be at a certain, fixed carrier frequency. Consequently, those methods are not suitable for automotive radar applications using linear frequency ramps. Therefore, it is necessary to develop an approach to estimate and monitor the PN PSD contained in a chirp signal, which will be the scope of the next section.

### PN estimation of FMCW signals

One of the most intuitive approaches to estimate the PN PSD from a linear FMCW signal would be to measure the PN PSD at multiple carrier frequencies covering the desired chirp bandwidth and form a simple average [24]. However, one disadvantage of this approach is that the PLL is operating in a different configuration as it is actually utilized in automotive applications. As a consequence, some disruptive effects may not occur and thus cannot be monitored.

In the following, we present two techniques to estimate the PN contained in an FMCW signal during operation [24]. The proposed methods were implemented on the hardware prototype presented in [17] and make use of the artificial OCT, which was also utilized to perform the SR leakage cancelation mentioned previously. For this application, the OCT is used as a BIST or design for testing. Similar to the DLD method presented in the “PN Measurement of CW Signals” section, the OCT is realized with a delay line. However, the DLD technique is not suitable for our problem at hand because the constant phase shift of  $90^\circ$  cannot be maintained for frequency-modulated input signals.

The system model with the artificial OCT is illustrated in Figure 7(b). Assuming the path that directly feeds the mixer has zero delay and with the Tx signal defined in (7), the OCT IF signal can be written as

$$s_{IF,O}(t) = A_{IF,O} \cdot \cos(2\pi f_{B,O}t + \Phi_O + \Delta\varphi_O(t)), \quad (9)$$

with  $f_{B,O}$  denoting the beat frequency corresponding to the time delay  $\tau_O$  caused by the OCT and  $\Delta\varphi_O(t)$ , the resulting DPN.

Applying the cosine sum identity and the small-angle approximation (the DPN is assumed to be sufficiently small), (9) can be rewritten as

$$s_{IF,O}(t) \approx A_{IF,O} \cdot \cos(2\pi f_{B,O}t + \Phi_O) - A_{IF,O} \cdot \sin(2\pi f_{B,O}t + \Phi_O) \Delta\varphi_{O,L}(t). \quad (10)$$

Note that the random phase term  $\Delta\varphi_O(t)$  from (9) turned into a random amplitude  $\Delta\varphi_{O,L}(t)$  due to this approximation. Because it is additionally affected by the LPF, it is referred to as *low-pass filtered DPN* and denoted with an additional subscript L in the sequel. In the following, we present the two mentioned methods based on (10), which are designated as PN estimation in the time and frequency domain, respectively.

### PN estimation in the time domain

In the first approach, the low-pass filtered DPN  $\Delta\varphi_{O,L}(t)$  is extracted in time domain by rearranging (10) as

$$\Delta\varphi_{O,L}(t) \approx \frac{A_{IF,O} \cdot \cos(2\pi f_{B,O}t + \Phi_O) - s_{IF,O}(t)}{A_{IF,O} \cdot \sin(2\pi f_{B,O}t + \Phi_O)}. \quad (11)$$

Because the design parameters  $A_{Tx}$ ,  $A_O$ ,  $\tau_O$ , and  $k$  are known, the amplitude of the IF signal  $A_{IF,O}$  and the beat frequency  $f_{B,O}$  as well as the constant phase term  $\Phi_O$  can be determined as shown in the section “Velocity Estimation.” To obtain the PSD  $S_{\Delta\varphi_{O,L}\Delta\varphi_{O,L}}(f)$  of the low-pass filtered DPN  $\Delta\varphi_{O,L}(t)$ , a conventional PSD estimation is applied. In this application, we are interested in the PSD of the PN  $\varphi(t)$  contained in the Tx signal, which is directly related to the unfiltered DPN PSD via the so-called range correlation term [25]. Accordingly, the PSD of the unfiltered DPN  $\Delta\varphi_O(t)$  can be calculated as

$$\frac{S_{\Delta\varphi_O\Delta\varphi_O}(f)}{\text{DPN PSD}} = \frac{S_{\varphi\varphi}(f)}{\text{PN PSD}} \underbrace{2(1 - \cos(2\pi f\tau_O))}_{\text{range correlation term}}. \quad (12)$$

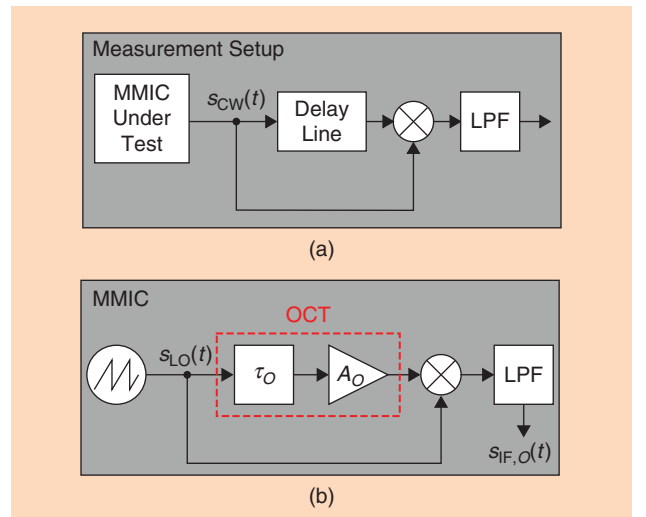
Including the magnitude response of the LPF  $|H_{LPF}(f)|$  into our considerations, based on (12), we obtain an expression for the PN PSD

$$S_{\varphi\varphi}(f) = \frac{S_{\Delta\varphi_{O,L}\Delta\varphi_{O,L}}(f)}{2 |H_{LPF}(f)|^2 [1 - \cos(2\pi f\tau_O)]}. \quad (13)$$

Although this method can be computed efficiently using the FFT to calculate  $S_{\Delta\varphi_{O,L}\Delta\varphi_{O,L}}(f)$ , it is possible to bypass the extraction process by directly calculating the PN PSD in the frequency domain, as explained next.

### PN estimation in the frequency domain

For the second technique, we reconsider the approximation from (10)



**FIGURE 7.** The system models describing the PN measurement system for (a) CW signals using a DLD approach as well as (b) estimating the PN PSD of a linear chirp utilizing the OCT.



$$s_{\text{IF},\text{O}}(t) \approx \underbrace{A_{\text{IF},\text{O}} \cos(2\pi f_{\text{B},\text{O}} t + \Phi_{\text{O}})}_{s_{\text{D}}(t)} - \underbrace{A_{\text{IF},\text{O}} \sin(2\pi f_{\text{B},\text{O}} t + \Phi_{\text{O}}) \Delta\varphi_{\text{O},\text{L}}(t)}_{s_{\text{R}}(t)}, \quad (14)$$

and we split it into a deterministic signal part  $s_{\text{D}}(t)$  and a random signal part  $s_{\text{R}}(t)$  containing the low-pass filtered DPN  $\Delta\varphi_{\text{O},\text{L}}(t)$ . In [24], it is derived in detail that the PSD of the random signal part can be evaluated as

$$S_{s_{\text{R}},s_{\text{R}}}(f,t) \approx A_{\text{IF},\text{O}}^2 S_{\varphi\varphi}(f) |H_{\text{LPF}}(f)|^2 [1 - \cos(2\pi f \tau_{\text{O}})] \gamma_{\text{R}}(t), \quad (15)$$

while the PSD of the deterministic part is described by

$$S_{s_{\text{D}},s_{\text{D}}}(f,t) \approx \frac{A_{\text{IF},\text{O}}^2}{2} \delta(f) \gamma_{\text{D}}(t) \quad (16)$$

with  $\delta(f)$  the Dirac delta function. Further,  $\gamma_{\text{R}}(t)$  and  $\gamma_{\text{D}}(t)$  are correction terms to compensate for the nonstationarity of the signals  $s_{\text{R}}(t)$  and  $s_{\text{D}}(t)$  and, thus, the time dependency of the PSDs.

As shown in [24], the PSD of the IF signal  $s_{\text{IF},\text{O}}(t)$  can be written as

$$S_{s_{\text{IF},\text{O}},s_{\text{IF},\text{O}}}(f,t) = S_{s_{\text{D}},s_{\text{D}}}(f,t) + S_{s_{\text{R}},s_{\text{R}}}(f,t). \quad (17)$$

Finally, inserting (15) into (17), the desired PN PSD can be expressed as

$$S_{\varphi\varphi}(f) \approx \frac{S_{s_{\text{IF},\text{O}},s_{\text{IF},\text{O}}}(f,t) - S_{s_{\text{D}},s_{\text{D}}}(f,t)}{A_{\text{IF},\text{O}}^2 |H_{\text{LPF}}(f)|^2 [1 - \cos(2\pi f \tau_{\text{O}})] \gamma_{\text{R}}(t)}. \quad (18)$$

In summary, to perform the PN estimation in the frequency domain, the PSD of the IF signal  $s_{\text{IF},\text{O}}(t)$  is estimated first. Then, this estimate in combination with the known design parameters  $A_{\text{Tx}}, A_{\text{O}}, \tau_{\text{O}},$  and  $k$  is used in (18) to determine the PSD of the PN contained in the Tx signal  $\varphi(t)$ . Both methods presented in this section require the integration of a delay line inside the MMIC, which is difficult to realize. Therefore, we will now present an alternative approach that enables PN estimation in cascaded radar system setups.

### PN estimation with FMCW signals in cascaded system configuration

As mentioned in the section “The FMCW Radar Principle,” the spatial resolution of an FMCW radar system is limited by the number of Tx and Rx antennas. One approach to increase the number of channels is to cascade multiple radar MMICs [26], [27]. Depending on the MMICs used, different approaches can be found to synchronize several MMICs to build up a cascaded system [28], [29]. For our novel PN PSD estimation technique, we make use of the cascaded system setup presented in [29]. In this cascaded system when operating in application mode, one master MMIC generates the LO signal and distributes it to the slave MMICs. The slave MMICs use the received LO signals both for transmission and for down-conversion of the received

radar signals. Figure 8(a) depicts an exemplary system configuration with one master MMIC and two slave MMICs acting in application mode.

In contrast, if the system is operating in our proposed PN PSD estimation mode [30], only two arbitrary MMICs are involved in the measurement simultaneously. These two MMICs are synchronized through a system clock. In this mode, both MMICs use their PLL to generate the FMCW signals  $s_1(t)$  and  $s_2(t)$ , which are defined as described in (7) using the same slope  $k$  but different starting frequencies  $f_1$  and  $f_2$ , respectively. As can be seen from Figure 8(b), the radio-frequency Tx output channel of the master MMIC is now reconfigured to be an input channel, driven by the slave MMIC. Finally, the signals  $s_1(t)$  and  $s_2(t)$  are mixed and low-pass filtered in the master MMIC.

According to the investigations previously described, the IF signal can be written (compare (10) and [30]) as

$$s_{\text{IF},\text{c}}(t) \approx A_{\text{IF},\text{c}} \cos(2\pi f_{\text{B},\text{D}} t + \Phi_{\text{D}}) - A_{\text{IF},\text{c}} \sin(2\pi f_{\text{B},\text{D}} t + \Phi_{\text{D}}) \varphi_{\text{D},\text{L}}(t), \quad (19)$$

where  $\Phi_{\text{D}}$  is a constant phase term composed of known design parameters and  $f_{\text{B},\text{D}}$  is the frequency difference of the FMCW signals  $s_1(t)$  and  $s_2(t)$  at the mixer, which is defined by

$$f_{\text{B},\text{D}} = f_1 - f_2 + kt_{\text{D}}. \quad (20)$$

The frequency component  $kt_{\text{D}}$  in (20) is caused by the fact that the mixing process is performed in the master, where  $t_{\text{D}}$  represents the difference in arrival time between the input signals of the mixer. Further, note that  $\varphi_{\text{D},\text{L}}(t)$  does not represent the DPN but rather the low-pass filtered version of the difference of the PN time domain signals  $\varphi_1(t) - \varphi_2(t - t_{\text{D}})$ . These two PN signals are uncorrelated because the signals  $s_1(t)$  and  $s_2(t)$  are generated by two different PLLs.

Assuming that  $t_{\text{D}}$  is known (e.g., as the result of an estimation procedure or by knowledge of the line lengths of the cascaded system), the aim of our approach is to set  $f_{\text{B},\text{D}}$  to zero. This is achieved by properly choosing the start frequencies  $f_1$  and  $f_2$  according to (20). With this, (19) simplifies to

$$s_{\text{IF},\text{c}}(t) \big|_{f_{\text{B},\text{D}}=0} \approx A_{\text{IF},\text{c}} \cos(\Phi_{\text{D}}) - A_{\text{IF},\text{c}} \sin(\Phi_{\text{D}}) \varphi_{\text{D},\text{L}}(t). \quad (21)$$

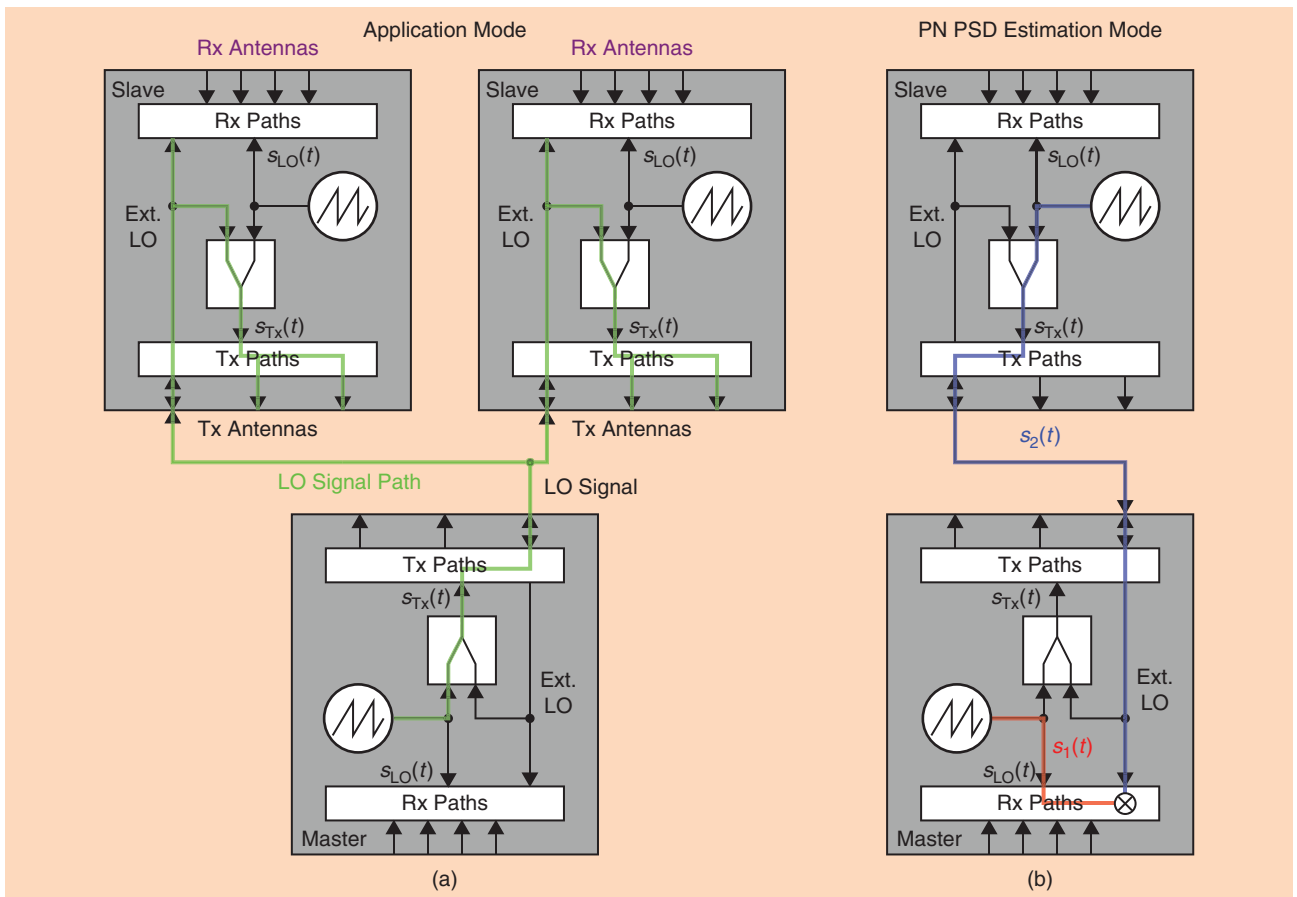
As shown in [30], the PSD of  $s_{\text{IF},\text{c}}(t)$  can be approximated with

$$S_{s_{\text{IF},\text{c}},s_{\text{IF},\text{c}}}(f) \approx A_{\text{IF},\text{c}}^2 \cos^2(\Phi_{\text{D}}) \delta(f) + 2A_{\text{IF},\text{c}}^2 \sin^2(\Phi_{\text{D}}) S_{\varphi_{\text{L}},\varphi_{\text{L}}}(f) \quad (22)$$

with  $S_{\varphi_{\text{L}},\varphi_{\text{L}}}(f)$  the mean PN PSD of the master and the slave MMIC. Finally, rearranging (22) yields this mean PN PSD

$$S_{\varphi_{\text{L}},\varphi_{\text{L}}}(f) = \frac{1}{2A_{\text{IF},\text{c}}^2 \sin^2(\Phi_{\text{D}})} S_{s_{\text{IF},\text{c}},s_{\text{IF},\text{c}}}(f) - \frac{\cos^2(\Phi_{\text{D}})}{2 \sin^2(\Phi_{\text{D}})} \delta(f). \quad (23)$$

This approach can be used to detect if one (or both) involved MMICs generate PN out of specification. To identify the erroneous MMIC, the cascaded system has to be composed of



**FIGURE 8.** The cascaded MMIC setup in application mode as well as in PN PSD estimation mode. (a) The exemplary system configuration with one master MMIC and two slave MMICs acting in application mode. (b) The system configuration for the PN PSD estimation mode with two arbitrary MMICs. Ext.: external.

at least three MMICs. Then the estimation process has to be repeated for every possible MMIC combination.

In contrast to the methods presented in the “PN Measurement of CW Signals” section, our novel approach enables PN monitoring using FMCW signals. As an advantage over the methods mentioned in the section “PN Estimation of FMCW Signals,” no delay line integration is required, but instead, it requires an underlying cascaded radar MMIC system. Further, the radar MMICs used have to support specific functionalities required for the PN PSD estimation mode.

## Conclusions

In this article, we illustrated the basics of FMCW radar signal processing, providing the most important formulas as well as supplementary examples. We demonstrated how to extract the range, relative velocity, and angle information from the measured IF signal, and we gave an insight into a common state-of-the-art radar signal processing technique in tutorial style. Further, we showed the impact of PN on radar measurement accuracy and pointed out the relevance of PN monitoring to verify that the PLL generates PN within its specifications. Because PN is the main figure of merit for a PLL, such monitoring functions are indispensable in terms of FuSa for ADAS and future AD. Finally, as the core of this article, state-of-the-art methods to

estimate the PSD of the PN contained in an FMCW signal in compliance to the ISO 26262 standard were presented.

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